Replacing brittle UI scripts with structured JSON & ETL pipelines

Instead of fragile XPath selectors that break whenever a button moves, we inspect underlying XHR/JSON network responses, validate every record with Pydantic schemas, and deliver clean Excel or database outputs.

19 sec	14,800	99.94%
Down from 42 min	Clean rows / run	Schema pass rate

Client Bottleneck Resolved: An operations team spent 6 hours weekly copying supplier catalog pricing across paginated SPA portals. Delivered a single-command Python 3.11 + Playwright pipeline with automatic retry backoff and styled XLSX reporting.

Runtime Telemetry — Playwright Async + Pydantic v2 Status: 0 Unhandled Exceptions

Stage 1
Session Context
Playwright async context loads dynamic SPA state cleanly.

Stage 2
JSON Capture
Structured XHR/Fetch payload captured with auto-retry.

Stage 3
Schema Guard
Strict type coercion, E.164 phone & SHA-256 dedup.

Stage 4
Clean Export
Styled multi-tab XLSX, UTF-8 CSV, or SQL/API upsert.

Sample Validated Output Preview (Deduplicated & Type-Checked)

record_id	supplier_sku	unit_price_usd	inventory_state	http_status	row_digest
REC-00481	IND-VALVE-88A	$142.50	In Stock (240)	200 VALID	9f84c2 ... e10b
REC-00482	IND-PUMP-310X	$389.00	Low Stock (12)	200 VALID	4a19d7 ... 7c32
REC-00483	IND-GAUGE-19M	$64.25	In Stock (890)	200 VALID	c702b5 ... 08fa
REC-00484	IND-SEAL-404K	$18.90	Restocking	200 VALID	2d51e9 ... 94a1

delivery_bundle/src/pipeline.py — Clean Modular Handover

```python
from pydantic import BaseModel, Field
from tenacity import retry, wait_exponential, stop_after_attempt

class CatalogItem(BaseModel):
    sku: str = Field(min_length=3)
    price_usd: float = Field(ge=0)
    in_stock: bool
    record_hash: str

@retry(
    wait=wait_exponential(multiplier=1, min=2, max=16),
    stop=stop_after_attempt(4)
)
async def fetch_page_batch(client, cursor: str) →
list[CatalogItem]:
    resp = await client.get("/api/v2/catalog", params={"cursor":
cursor})
    resp.raise_for_status()
    return [CatalogItem.model_validate(r) for r in resp.json()
["items"]]
```

```
[OK] Exported 14,800 validated rows → output/catalog_clean.xlsx (0
duplicates)
```

Every delivery is packaged so non-coders can run it in one click

No undocumented spaghetti scripts. You receive a clean, self-contained folder with environment templates and verified output datasets.

01 / Modular Source Code
Clean Python 3.11+ files with clear inline comments and error logs.

02 / Config Without Coding
Simple `.env.example` & config file to change inputs anytime.

03 / Quick-Start README
Copy-paste setup commands for Windows, macOS, Linux, or Docker.

04 / Verified Output File
Styled Excel (`.xlsx`), UTF-8 CSV, JSON, or live API/Sheet sync.

Select Your Package Tier (Or Message First for a Custom Quote)

Basic — 1 Day	Standard — 2 Days	Premium — 4 Days
$20	**$65**	**$145**
Starter Script or PoC	**Pro Extractor & API Sync**	**Full Automated ETL**
Single-source workflow or extraction script + clean CSV/XLSX output.	Multi-page dynamic SPA extraction or REST API sync + auto-retries.	Scheduled end-to-end pipeline, DB/CRM upsert, proxy config & Docker.